the object. An effective CTF can also be calculated directly from simulated STEM images where both the object and the image are known. We use this fact to validate our model. Part of the expression we derive here was already derived in Hawkes & Kasper [6] but without the quadratic term which, as we will prove in this work, is describing the ADF contribution. Starting from the derivation in [6], which, as we will demonstrate, describes (A)BF-STEM, we develop a description that includes both bright field and dark field imaging.

One of the well-known results of the theoretical work up till now is that an analytical expression for the CTF in the limiting case of a BF detector consisting of a point detector centered on the optical axis was found (named BF_δ here). This was achieved using the reciprocity theorem [12–15] which shows that it is equivalent to the case of TEM for which a CTF was derived earlier. As stated by Rose [12] “this equivalence [between CTEM and STEM] has been pointed out in the past on numerous occasions” (e.g. [14,15]). A mathematical proof based on the Bloch-wave formalism was given in [13]. We include an alternative mathematical proof that is valid for any type of thin sample not involving the Bloch-wave formalism.

Also in the case of ADF-STEM an approximate formula was established [1,2,6,9] based on the notion that in this case the imaging can be regarded as being incoherent. This led to an effective description of ADF and high angle annular dark field (HAADF) imaging, without a rigorous mathematical proof.

The main issue in the derivation of a simple, single formula STEM model has always been that a detector measures the *square* of the electron wave function at the detector (although this is not the fundamental reason that STEM is nonlinear). This gives rise to a quadratic term which is difficult to deal with mathematically. Many authors therefore limited their analysis to only the first order (or in some cases the second order) of the Taylor expansion of the sample transmission function. Hawkes & Kasper [6] work out the theory for arbitrary order in the sample transmission function but are left with a quadratic term that “cannot be simplified in any useful way”. This quadratic term is, however, essential in describing DF imaging and cannot be ignored. In our derivation we get an equivalent result, formulated in a more operational way, but this formulation allows a further investigation of this quadratic term, thereby linking the expressions for BF and DF imaging into one overall expression.

Within the framework of our derivation there is no need to impose conditions on the sample, such as the weak phase object approximation or the presence of a lattice, as is often encountered in the literature [1,5,13,16–19]. Both crystalline and amorphous samples can be treated equally well using our expression.

The nonlinearity of STEM imaging requires careful consideration of what is to be considered the object that is imaged. Conventionally the desired object is the phase of the output wave after transmission through the sample. The nonlinearity prevents us from obtaining a sample-independent CTF when the object is defined as the phase of the output electron wave. Instead, we obtain an expression concerning two different aspects of the sample. The first aspect could be considered as the object in (A)BF for weakly scattering samples and the second as the object for (HA)ADF imaging. For both of these special cases an effective (linear) CTF is found. These effective CTFs are already known in the literature, but often without the precise definition of the corresponding object.

In Section 2 we start out with a mathematical formulation of the STEM imaging process, and introduce our notation and assumptions. A proof of the reciprocity theorem is included as well as a note on the nonlinearity of the STEM imaging process. The simulation model that is used to validate our theoretical model is described in Section 3. The central equation describing an STEM image is derived in Section 4. The final step of obtaining the effective CTF for STEM is then taken in Section 5. In the same section we show the two cases which correspond to the parts of the theory that have already been found in the literature. The first corresponds to the well-known weak phase object approximation (WPA), the second to pure dark field imaging. We prove that in both cases our derived general formula reduces to the well-known formulas for BF_δ and (HA)ADF. Furthermore we briefly outline a recipe to include incoherencies of the source and thermal fluctuations of the sample in the model if required. Finally we present our conclusions in Section 6.

2. Mathematical formulation of the STEM imaging process

The STEM imaging process is schematically shown in the left part of Fig. 1. In the first stage a probe is formed using

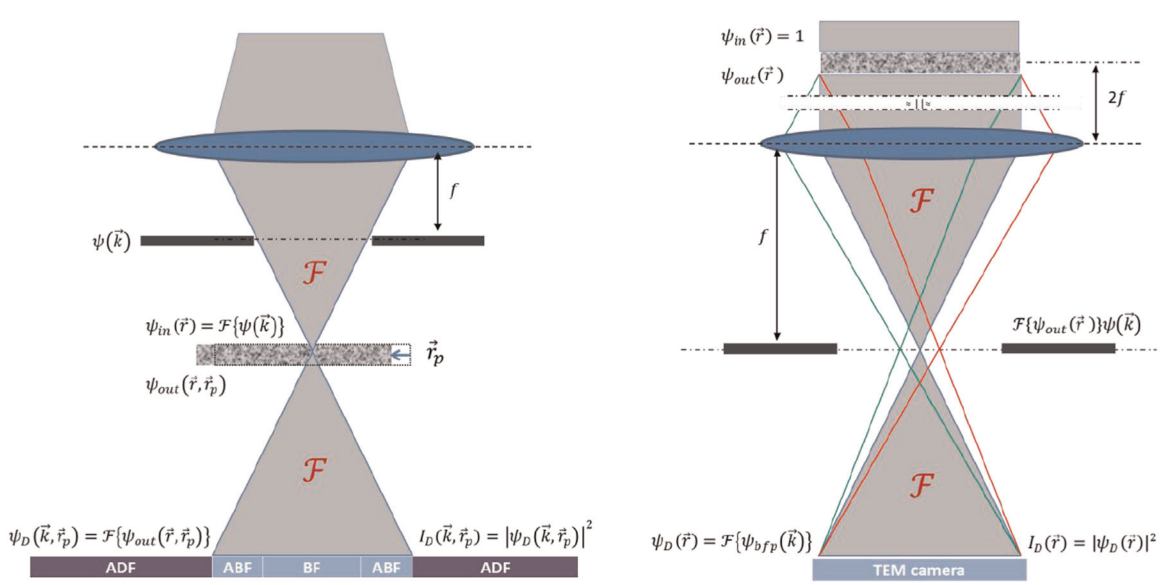

Fig. 1. Schematic overview of the STEM imaging process (left) and for comparison the TEM imaging process (right). The strong black lines indicate the aperture, denoted with $A(\vec{k})$, which is included in $\psi(\vec{k})$.

only its squared amplitude. The signal in the detection plane is therefore

$$I_D(\vec{k}, \vec{r_p}) = \left|\psi_D^{STEM}(\vec{k}, \vec{r_p})\right|^2 = \psi_D^{STEM}(\vec{k}, \vec{r_p})\cdot\overline{\psi_D^{STEM}(\vec{k}, \vec{r_p})} \tag{13}$$

where the notation $\bar{x}$ indicates the complex conjugate of x.

In order to obtain the final STEM image, the intensity distribution in the detector plane must be integrated over the active area of a specific detector in that plane. This corresponds to integrating (13) over a range of wave numbers $\vec{k}$. The selection of this range of wave numbers is realized using the detector window function $W(\vec{k})$ as introduced in (9). The STEM image is given by

$$\begin{aligned} I^{STEM}(\vec{r_p}) &= \iint_{-\infty}^{\infty} W(\vec{k}) I_D(\vec{k}, \vec{r_p}) d^2\vec{k} \\ &= \iint_{-\infty}^{\infty} W(\vec{k}) \left|\psi_D^{STEM}(\vec{k}, \vec{r_p})\right|^2 d^2\vec{k}. \end{aligned} \tag{14}$$

Writing out this integral we obtain

$$\begin{aligned} I^{STEM}(\vec{r_p}) = \iint_{-\infty}^{\infty}\iint_{-\infty}^{\infty}\iint_{-\infty}^{\infty} \left(1 - \zeta(\vec{r} + \vec{r_p})\right)\left(1 - \overline{\zeta(\vec{r}\,' + \vec{r_p})}\right)\psi_{in}(\vec{r}) \\ \overline{\psi_{in}(\vec{r}\,')}\, W(\vec{k}) e^{-2\pi i\vec{k}\cdot(\vec{r} - \vec{r}\,')} d^2\vec{r} d^2\vec{r}\,' d^2\vec{k} \end{aligned} \tag{15}$$

Working out the parentheses involving ζ we obtain four terms which can be rewritten as

$$\begin{aligned} I^{STEM}(\vec{r_p}) &= \iint_{-\infty}^{\infty} W(\vec{k}) \left|\mathcal{F}\left\{\psi_{in}(\vec{r})\right\}(\vec{k})\right|^2 d^2\vec{k} \\ &- \iint_{-\infty}^{\infty} \zeta(\vec{r} + \vec{r_p})\cdot\psi_{in}(\vec{r})\mathcal{F}\left\{W(\vec{k})\mathcal{F}^{-1}\left\{\overline{\psi_{in}(\vec{r}\,')}\right\}(\vec{k})\right\}(\vec{r}) d^2\vec{r} - \iint_{-\infty}^{\infty} \overline{\zeta(\vec{r}\,' + \vec{r_p})}\cdot\overline{\psi_{in}(\vec{r}\,')}\mathcal{F}^{-1}\left\{W(\vec{k})\mathcal{F}\left\{\psi_{in}(\vec{r})\right\}(\vec{k})\right\}(\vec{r}\,') d^2\vec{r}\,' \\ &+ \iint_{-\infty}^{\infty}\iint_{-\infty}^{\infty} \zeta(\vec{r} + \vec{r_p})\overline{\zeta(\vec{r}\,' + \vec{r_p})}\psi_{in}(\vec{r})\overline{\psi_{in}(\vec{r}\,')}\mathcal{F}\left\{W(\vec{k})\right\}(\vec{r} - \vec{r}\,') d^2\vec{r} d^2\vec{r}\,' \end{aligned} \tag{16}$$

where we have executed the integration over $\vec{r}$ and $\vec{r}\,'$ in the first term, over $\vec{r}\,'$ and $\vec{k}$ in the second term, over $\vec{r}$ and $\vec{k}$ in the third term and over $\vec{k}$ in the last term. Note that the third term in (16) is the complex conjugate of the second term.

The first term in (16) is independent of $\vec{r_p}$ and hence a constant. It represents a DC offset of the STEM image.

Introducing a short-hand notation $B(\vec{r}) \equiv \psi_{in}(\vec{r})\mathcal{F}\left\{W(\vec{k})\mathcal{F}^{-1}\left\{\overline{\psi_{in}(\vec{r}\,')}\right\}(\vec{k})\right\}(\vec{r})$ we can write the second and third terms of (16) as

$$\begin{aligned} &- \iint_{-\infty}^{\infty} \left(\zeta(\vec{r} + \vec{r_p})\cdot B(\vec{r}) + \overline{\zeta(\vec{r} + \vec{r_p})}\cdot\overline{B(\vec{r})}\right) d^2\vec{r} = \\ &- \iint_{-\infty}^{\infty} \frac{\zeta(\vec{r} + \vec{r_p}) + \overline{\zeta(\vec{r} + \vec{r_p})}}{2}\cdot\left(B(\vec{r}) + \overline{B(\vec{r})}\right) d^2\vec{r} \\ &- \iint_{-\infty}^{\infty} \frac{\zeta(\vec{r} + \vec{r_p}) - \overline{\zeta(\vec{r} + \vec{r_p})}}{2i}\cdot i\left(B(\vec{r}) - \overline{B(\vec{r})}\right) d^2\vec{r} \end{aligned} \tag{17}$$

where it should be noted that the combinations $B(\vec{r}) + \overline{B(\vec{r})}$ and $i\left(B(\vec{r}) - \overline{B(\vec{r})}\right)$ are both real.

Using (17) and (11), we rewrite (16) as

$$\begin{aligned} I^{STEM}(\vec{r_p}) = &\iint_{-\infty}^{\infty} W(\vec{k})\left|\mathcal{F}\{\psi_{in}(\vec{r})\}(\vec{k})\right|^2 d^2\vec{k} + \\ &\iint_{-\infty}^{\infty} \sin\varphi(\vec{r} + \vec{r_p})\cdot i\left(B(\vec{r}) - \overline{B(\vec{r})}\right) d^2\vec{r} + \\ &- \iint_{-\infty}^{\infty} (1 - \cos\varphi(\vec{r} + \vec{r_p}))\cdot\left(B(\vec{r}) + \overline{B(\vec{r})}\right) d^2\vec{r} + \\ &\iint_{-\infty}^{\infty}\iint_{-\infty}^{\infty} \zeta(\vec{r} + \vec{r_p})\overline{\zeta(\vec{r}\,' + \vec{r_p})}\psi_{in}(\vec{r})\overline{\psi_{in}(\vec{r}\,')}\mathcal{F}\{W(\vec{k})\}(\vec{r} - \vec{r}\,') d^2\vec{r} d^2\vec{r}\,'. \end{aligned} \tag{18}$$

Note that the second and third terms in (18) have the form of a cross-correlation integral[3]. Also note that the third and fourth terms are of order φ^2 and higher whereas the second term is of order φ and higher.

Next, we proceed with the last term in (18), which, after introducing a change of variable $\vec{r}\,'' = \vec{r} - \vec{r}\,'$, reads

$$\begin{aligned} I_{\zeta\bar{\zeta}}(\vec{r_p}) \equiv \iint_{-\infty}^{\infty}\iint_{-\infty}^{\infty} \zeta(\vec{r} + \vec{r}\,'' + \vec{r_p})\overline{\zeta(\vec{r} + \vec{r_p})}\psi_{in}(\vec{r} + \vec{r}\,'')\overline{\psi_{in}(\vec{r})} \\ \mathcal{F}\left\{W(\vec{k})\right\}(\vec{r}\,'') d^2\vec{r}\,'' d^2\vec{r}. \end{aligned} \tag{19}$$

Using the definition of the cross-correlation (19) can be rewritten in terms of the autocorrelation function of the product $\zeta(\vec{r} + \vec{r_p})\psi_{in}(\vec{r})$ as

$$\begin{aligned} I_{\zeta\bar{\zeta}}(\vec{r_p}) = \iint_{-\infty}^{\infty} \left(\zeta(\vec{r} + \vec{r_p})\psi_{in}(\vec{r})\star\zeta(\vec{r} + \vec{r_p})\psi_{in}(\vec{r})\right)(\vec{r}\,'') \\ \cdot\mathcal{F}\left\{W(\vec{k})\right\}(\vec{r}\,'') d^2\vec{r}\,''. \end{aligned} \tag{20}$$

The autocorrelation function of a coherent probe ψ_{in} (i.e. excluding the effects of incoherence of the source) is given by $\left(\psi_{in}(\vec{r})\star\psi_{in}(\vec{r})\right)(\vec{r}\,'') = \mathcal{F}\left\{A^2(\vec{k})\right\}(\vec{r}\,'')$, which directly follows from the Wiener–Khinchin theorem, and drops to zero over a length scale of the order of $1/k_{BF}$. The autocorrelation function of the sample typically also shows an initial drop (to a finite constant value) as a function of $\vec{r}\,''$ over a distance of the size of an atom. For periodic samples it will peak again at positions corresponding to the lattice periodicity, for amorphous samples this autocorrelation will always remain small for large $\vec{r}\,''$. The autocorrelation of the product of the two functions also drops to zero and stays at zero, provided that the

[3] The cross-correlation is defined as $\left(f(\vec{r}\,')\star g(\vec{r}\,')\right)(\vec{r}) \equiv \iint_{-\infty}^{\infty} f(\vec{r}\,')g(\vec{r} + \vec{r}\,')d^2\vec{r}\,'$.